

The Three practice exercises at the last of slide section one

```

1  import 'dart:io';
2
3  Run | Debug
4  void main() {
5      /*Exercise 1: Email Validator
6       Create a Dart program that validates whether a given string is a valid email address.
7       Use string methods to check for the presence of the "@" symbol,
8       proper domain format, and other email address criteria.*/
9
10     print("Enter your email:");
11     String email = stdin.readLineSync();
12     bool contAt = email.contains('@');
13     if (contAt) {
14         print("Your email is valid");
15     } else {
16         print('Your email is not valid');
17     }
18 }
19
20 Exercise 2: Sentence Reversal
21 Write a Dart program that takes a sentence as input and reverses the order of words in it.
22 For example, if the input is "Hello World," the program should output "World Hello."
23 These exercises will help you practice various string methods in Dart, such as splitting,
24 reversing, and checking for specific patterns, while also improving your problem-solving skills.*/
25
26 print("Enter a sentence to be reversed");
27 String notRev = stdin.readLineSync();
28 String rev = "";
29 for (int i = notRev.length - 1; i >= 0; i--) {
30     rev = rev + notRev[i];
31 }
32 print(rev);
33 print("////////////////////////////////////////");
34
35 /*Exercise 3: Uppercase and Lowercase Conversion
36 Write a Dart program that takes a string as input and converts it to uppercase and lowercase.
37 Print both the uppercase and lowercase versions of the string.*/
38
39 print(" The sentence:");
40 String Orig = stdin.readLineSync();
41 String upper = Orig.toUpperCase();
42 String lower = Orig.toLowerCase();
43 print("The Upper Case : $upper\nThe Lower Case : $lower");
44 print("////////////////////////////////////////");
45
46
47
48

```

```

3 void main() {
10     bool contAt = email.contains('@');
11     if (contAt) {
12         print("Your email is valid");
13     } else {
14         print('Your email is not valid');
15     }
16     print("////////////////////////");
17     /*
18 Exercise 2: Sentence Reversal
19 Write a Dart program that takes a sentence as input and reverses the order of words in it.
20 For example, if the input is "Hello World," the program should output "World Hello."
21 These exercises will help you practice various string methods in Dart, such as splitting,
22 reversing, and checking for specific patterns, while also improving your problem-solving skills.*/
23     print("Enter a sentence to be reversed");
24     String notRev = stdin.readLineSync();
25     String rev = "";
26     for (int i = notRev.length - 1; i >= 0; i--) {
27         rev = rev + notRev[i];

```

[illegible]

The assignment of section 1

```
AssignmentS1.dart > main
1  import 'dart:io';
2  import 'dart:math';
3
4  void main() {
5      /*Exercise 1: Calculate Circle Area
6       Write a Dart program that calculates the area of a circle.
7       Take the radius as input and print the area.*/
8
9      print("////////////////////////////////////////");
10     print("Calculate Circle Area\n  Enter the radius: ");
11     double radius = double.parse(stdin.readLineSync());
12     double area = 3.14 * pow(radius, 2);
13     print("Area is: $area");
14     print("////////////////////////////////////////");
15
16     /* Exercise 2: Check Odd or Even
17     Write a program that checks if a given number is odd or even. Take a
18     number as input and print whether it's odd or even.*/
19
20     print("////////////////////////////////////////");
21     print("To check even or odd\n  Enter the number: ");
22     int num = int.parse(stdin.readLineSync());
23     if (num % 2 == 0) {
24         print("The number you entered is Even!");
25     } else {
26         print("The number you entered is Odd!");
27     }
28     print("////////////////////////////////////////");
29 }
```

```
AssignmentS1.dart > main
4  void main() {
30     /* Exercise 3: Calculate BMI (Body Mass Index)
31     Write a program that calculates BMI, Take the weight (in kilograms)
32     and height (in meters) as input and print the BMI along with a message
33     indicating whether the person is underweight normal weight overweight,
34     or obese.*/
35
36     print("////////////////////////////////////////");
37     print("This is the BMI calculator\nEnter your wieght in kilogram: ");
38     int weight = int.parse(stdin.readLineSync());
39     print("\n");
40     print("Enter your height in meters: ");
41     double height = double.parse(stdin.readLineSync());
42     print("\n");
43     double BMI = weight / pow(height, 2);
44
45     if (BMI < 18) {
46         print("You are Underweight");
47     } else if (BMI < 25) {
48         print("You are okay");
49     } else {
50         print("You are overweight!");
51     }
52     print("////////////////////////////////////////");
53     /* Exercise 4: Convert Minutes to Hours and Minutes
54     Develop a program that converts a given number of minutes into hours
55     and minutes. Take the minutes as input and print the equivalent hours and
56     remaining minutes.*/
57     print("Convert Minutes to Hours and Minutes\n1: Minutes\n2: Hours");
58
59     int choice = int.parse(stdin.readLineSync());
60     print("Enter number: ");
61     double numOf = double.parse(stdin.readLineSync());
62 }
```

```
AssignmentS1.dart > main
4 void main() {
51 }
52 print("////////////////////////////////////////");
53 /* Exercise 4: Convert Minutes to Hours and Minutes
54 Develop a program that converts a given number of minutes into hours
55 and minutes. Take the minutes as input and print the equivalent hours and
56 remaining minutes.*/
57 print("Convert Minutes to Hours and Minutes\n1: Minutes\n2: Hours");
58
59 int choice = int.parse(stdin.readLineSync());
60 print("Enter number: ");
61 double numOf = double.parse(stdin.readLineSync());
62
63 if (choice == 1) {
64   print("This is ${numOf / 60} hours");
65 } else if (choice == 2) {
66   print("This is ${numOf * 60} minutes");
67 }
68 print("////////////////////////////////////////");
69
70 /* Exercise 5: Swap Two Numbers
71 Create a program that swaps the values of two variables. Take two
72 numbers as input, swap their values, and print the swapped values.
73 */
74 print("Swap two numbers\nEnter first:");
75 int a = int.parse(stdin.readLineSync());
76 print("Enter second:");
77 int b = int.parse(stdin.readLineSync());
78 int c = a;
79 a = b;
80 b = c;
81
82
```

```
////////////////////////////////////////
Calculate Circle Area
Enter the radius:
5
Area is: 78.5
////////////////////////////////////////
To check even or odd
Enter the number:
25
The number you entered is Odd!
////////////////////////////////////////
This is the BMI calculator
Enter your wieght in kilogram:
70

Enter your height in meters:
180

You are Underwight
```

```
You are Underwight
////////////////////////////////////////
Convert Minutes to Hours and Minutes
1: Minutes
2: Hours
1
Enter number:
60
This is 1.0 hours
////////////////////////////////////////
Swap two numbers
Enter first:
5
Enter second:
7
PS C:\Users\karam\OneDrive\Desktop\Course>
```